Digital Twins for Cyber-Resilient Power Grids: A Systematic Review and a Research Agenda for the AI-Powered Era

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Abstract

Digital twins are becoming the reference architecture for monitoring, protecting and controlling modern power grids, and artificial intelligence (AI) is being adopted inside those twins at speed. We review whether the security literature treats AI as a defensive tool faster than it models AI-enabled threats, or the twin itself, as an attack target. Following PRISMA 2020 style, we screened 1,287 unique OpenAlex records and retained 100 studies on digital twins for power-grid cybersecurity (2020–2026); 83 were coded at full text across 46 fields and 17 at abstract level. The review is restricted to power-grid papers that name a digital twin and a security term. AI is used defensively in 70 of 83 full-text studies, while 35/83 (42.2%) engage any AI-enabled threat and 11/83 (13.3%) engage one substantively; 48 (57.8%) do not engage at all. The twin itself is treated as an attack target in at most one study under a strict codebook definition (a vehicle-twin review). Threat coverage, as coded, concentrates at operation (66/83), with one design-phase study and none at commissioning ($\kappa = 0.32$). Two coders agreed on 35/83 at $\kappa = 0.60$ for "any engagement" and one of six at $\kappa = 0.14$ for "substantive" engagement; the results are bounds, not population estimates. We close with a five-item research agenda in which every item is tied to the gap it addresses and the evidence that would confirm it.

Keywords: digital twin, power grid, cyber-physical security, adversarial machine learning, systematic review, research agenda

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1. Introduction

Electric power grids are digitising faster than they are being secured. Wide-area measurement, distributed energy resources, electric-vehicle charging and market-coupled control have turned the grid into a cyber-physical system whose behaviour is increasingly mediated by software [1, 2]. The digital twin—a synchronised, bidirectional virtual counterpart of a physical asset, updated from live sensor streams—has moved in the same decade from concept to a reference architecture for grid monitoring, protection and control [3, 4]. That same synchronisation is what makes the twin valuable and what makes it a target.

Artificial intelligence (AI) has followed the twin into the grid. Machine learning now performs state estimation, anomaly detection, load and attack forecasting, and increasingly control [5, 6]. The security community has been quick to adopt it. The question this review asks is whether the same community has been as quick to model AI—and the twin itself—as something an adversary can attack. Throughout, 'digital twin' must be named by the authors, so adversarial-ML work on grid security that does not mention a twin is out of scope and is not represented in any figure here.

Early reading suggested a specific, testable asymmetry: grid digital-twin security research adopts AI as a defensive tool far faster than it models AI, or the twin, as an attack target. This review quantifies that asymmetry across a screened corpus and turns it into a research agenda. It is a theory-testing systematic review, not an empirical deployment study.

1.1. Positioning against prior reviews

The closest work, and the reason this review had to be repositioned, is Habeeb, Weli and Almaliki's 2026 PRISMA review of "Dual-Role Digital Twins" in cyber-physical security [7]. Their contribution is real and prior: they name the dual role—the twin as both an expanded attack surface and a defence enabler—and set out four research directions around it. Their corpus is 20 studies, gathered with a query that has no specific attack operator, and spread across manufacturing, healthcare, IIoT, smart city and water management, with the power grid represented in roughly one study.

We therefore do not claim to have discovered the dual-role tension. Our position is narrower and complementary. Habeeb et al. name two axes; we

test a narrower, grid-side instance of the attack-surface axis—with a published corpus, a closed threat vocabulary, and reported coder reliability, but only where a grid twin is explicitly an attack target. In other words, they build the theory; we report a lower bound on its grid manifestation, not a test of the full axis. Section 3 sets out the mapping in full, including the five attack vectors their review identifies and our corpus's coverage of them.

1.2. Contributions

1. A PRISMA-2020-style corpus of 100 studies on digital twins for power-grid cybersecurity, screened from 1,287 unique records with a query set that includes explicit adversarial-AI probes (Section 4).
2. A 46-field extraction of that corpus, coded by computer-assisted human reading, with the extraction method disclosed rather than hidden (Section 4).
3. A two-coder reliability study on a 20% blind sample, reported field by field including the fields where agreement is only fair (Section 4.8).
4. The headline measurement: AI is defensive in 70/83 studies; 35/83 engage any AI-enabled threat and 11/83 engage one substantively, while the twin is explicitly named an attack target in one study (Section 5.5).
5. A lifecycle concentration result: under a single-assigned lifecycle label, 66/83 studies sit at the operation phase (Section 5.4).
6. A mapping from our independent threat vocabulary to the five vectors of Habeeb et al. [7], showing coverage (vocabulary-level) plus six additional terms that their query could not return (Section 3); this is coverage, not evidence that the vectors are dealt with empirically.
7. A five-item research agenda in which every item is tied to the gap that justifies it and the evidence that would validate it (Section 7).

1.3. Research questions

RQ1 How is the digital twin positioned in grid cybersecurity research—as a defensive tool, an attack target, or both?

RQ2 Which threats are named, at which grid domains and lifecycle stages, and with what threat models?

RQ3 How far does the corpus engage AI-enabled threats (adversarial machine learning, evasion, poisoning, LLM-specific, agentic)?

RQ4 How rigorous is the evidence base—evaluation method, maturity, baselines, metrics, and released code or data?

RQ5 What are the resulting gaps, and what would close them?

1.4. Structure

Section 2 gives the technical background. Section 3 reviews prior work and positions this review against it. Section 4 is the protocol. Section 5 reports results, figure-led, across RQ1–RQ4. Section 6 discusses them. Section 7 sets the research agenda for RQ5. Section 8 is the threats-to-validity analysis, and Section 9 concludes.

2. Background

2.1. Digital twins in power systems

A digital twin is a virtual representation of a physical entity that is connected to it by a bidirectional data link, such that the virtual state tracks the physical state and, in the stronger formulations, the physical entity can be actuated through the twin. Three properties distinguish a twin from an ordinary model or simulation: synchronisation with live data, bidirectionality, and asset specificity. Grid applications span state estimation and observability, protection and control, fault and attack diagnosis, asset-health monitoring, and training and planning [3–5].

The maturity of grid twins varies widely. Some reported twins are conceptual architectures with no implementation; many are prototypes demonstrated on simulation or testbed data; a small number report validation against real grid assets [6, 8]. This maturity gradient matters for security, because a threat model that is plausible on a conceptual twin may not be reachable on a real deployment, and vice versa.

2.2. Cyber threats to grids and to twins

The classical grid threat model is adversarial and studied: false data injection against state estimation, replay, denial of service, man-in-the-middle on telemetry links, false command injection, and supply-chain compromise [1, 2]. These are conventional cyber-physical attacks and are well represented in the corpus (Section 5.4).

The twinning layer adds attack surface that the classical model does not cover. The bidirectional link can be exploited in both directions; the twin

synchronises with sensor streams, so any data-poisoning channel into those streams reaches the twin; and where the twin uses learned components—detectors, forecasters, controllers—those components can themselves be attacked by distribution-shift, evasion, or training-time poisoning. This is the attack-surface axis that Habeeb et al. name [7].

2.3. AI on both sides of the boundary

AI enters grid security from two directions. Defensively, machine learning performs intrusion and anomaly detection, attack classification, load and attack forecasting, and increasingly control. Offensively, the same model families are themselves vulnerable: adversarial examples can evade learned detectors, poisoning can corrupt training data drawn from compromised sensor streams, and large language models introduce prompt-level and social-engineering vectors that did not exist in earlier grids. An "AI-enabled threat" in this review means an attack in which the adversary uses AI, or attacks an AI component, or both—as distinct from a conventional cyber-physical attack that happens to be detected by an AI method.

A brief terminology note. 'Digital twin' has largely superseded 'CPS model' and 'HIL simulation' in recent grids literature, so papers that functionally twin a grid without using the phrase are out of scope of the search, despite carrying relevant threat models.

3. Related work and positioning

3.1. The closest review

Habeeb, Weli and Almaliki (2026, *Digital Twin*) review Dual-Role Digital Twins for cyber-physical security with PRISMA 2020: 412 identified, 318 screened, 72 eligibility-checked, 20 included, spanning January 2019 to December 2025 [7]. They ground the dual role in Complex Adaptive Systems, functional duality, and zero-trust architecture, and they name five attack vectors. Their search string is `("Digital Twin" OR "DT") AND ("Cybersecurity" OR "Attack Surface" OR "Defense" OR "Honeypot" OR "Intrusion Detection")`. It contains no term for a specific attack (no adversarial machine learning, poisoning, evasion, or reverse-pathway operator), so attack-side records have to enter through the security and defence terms. The same limitation applies to our twelve core queries, which is why we added the four adversarial-AI probes.

Table 1: Our independent threat vocabulary mapped onto the five attack vectors of Habeeb et al. [7]. Counts are records naming the term (full text, $n = 83$).

#	Their vector	Our vocabulary terms	Records naming it
i	Bidirectional data-link exploitation	`mitm`, `replay`, `dos`	26, 23, 49
ii	Adversarial data poisoning on sensor streams	`poisoning`, `fdi`	22, 46
iii	AI/ML model exploitation via training-distribution attacks	`ai_adv_ml`, `ai_poisoning`, `ai_evasion`	24, 19, 9
iv	Reverse attack pathway (twin → plant)	`reverse_pathway`, `false_command`	10, 22
v	Insider and supply-chain compromise of the twin	`insider`, `supply_chain`	20, 19

3.2. The bridge: their vectors, our vocabulary

Our codebook's threat vocabulary was designed independently, before we read their paper. The threat vocabulary, defined and reported in Section 5.4, predates this paper. Table 1 shows that our vocabulary covers all five of their attack vectors and adds terms that their query does not name, hence could not have returned directly: adversarial machine learning (24), LLM threats (4), agentic automation (2), reverse pathway (10), evasion (11), and a separate supply-chain term (19), in part because their corpus is small and not grid-focused and in part because their query has no specific attack-by-name operator.

3.3. Other reviews in the corpus

Several corpus entries are themselves reviews [1–5, 9]. Their coverage is broad on the defensive side. Our contribution relative to them is the quantified attack-side measurement: we do not merely note that AI threats matter, we count how often the corpus engages them and find that the answer is rarely, and rarely substantively.

4. Methodology

4.1. Design

This is a systematic literature review reported in PRISMA 2020 style on an OpenAlex-derived screened-and-selected subset. The protocol was fixed before extraction, with the codebook revisions noted below. The review is a theory-testing review: the asymmetry stated in the abstract's scope sentence is the hypothesis, and the retained 100 studies are the test.

4.2. Search strategy

We searched OpenAlex with twelve core queries combining a digital-twin term, a grid-domain term, and a security term, yielding 1,287 unique records after de-duplication. We then ran four additional adversarial-AI probes—combining digital-twin and grid terms with adversarial machine learning, poisoning, evasion, and large-language-model terms—which raised the candidate pool to approximately 2,000. The probes exist because the core grid-security vocabulary is not expected to retrieve the attack-side literature, and we wanted the asymmetry result to be robust to that retrieval bias.

4.3. Selection and inclusion

We screened on title and abstract and applied the eligibility rule in Section 4. The eligible set (492 records) was then narrowed to 100 by a non-probabilistic rule combining OpenAlex relevance score, citation count, and recency, with the final cut at rank 100. The released notebook reports per-query membership for the candidate pool (approximately 2,000 records retrieved by four adversarial-AI probe queries), but probe-only membership among the retained 100 is not summarised in it (for an honest reading, treat the probe share as zero unless shown); see Section 8, Search bias. Because the rule is not a random sample, every prevalence figure in Section 5 is a statement about the retained 100 studies and should not be read as an estimate of the wider literature. Wilson 95% intervals are shown for the Section 5.4 figures; elsewhere, small denominators bound all percentages.

4.4. Screening

Screening was title-and-abstract. A record was eligible if its title or abstract contained at least one digital-twin term, at least one grid or power or energy term, and at least one cyber or security term. Screening excluded 795 records missing at least one of the three terms (digital twin, grid or energy,

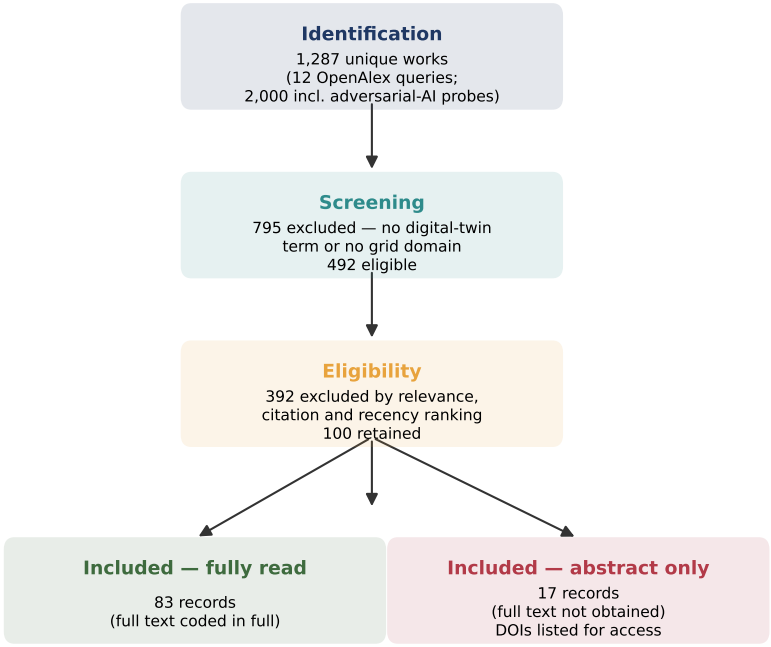


Figure 1: PRISMA-style flow. Eighty-three records are coded at full text and 17 at abstract level only, because their full text was not obtained.

cyber or security), leaving 492 eligible records. Inclusion then ranked the eligible set by relevance, citation count, and recency, and retained the top 100. Figure 1 is the flow.

4.5. Extraction

We extracted 46 fields per record. Extraction was computer-assisted but human-coded: a script surfaced keyword hits and context snippets for each field, and every code was assigned by a human reading those snippets in context. Keyword snippets were scored against the full text and the surrounding paragraph, not the excerpt alone; reference lists and author biographies were stripped before scanning, so absence claims rest on the surviving body text. The extraction is reproducible from the coded excerpts and DOIs, which are released.

Of the 100 records, 83 were available in full text and coded in full; 17 were available only as abstracts and were coded at abstract level only. All

abstract-only fields that require full text are treated as missing, not as zero, throughout. The 17% unobtainable rate and its direction are reported as a reproducibility finding in Section 8.

4.6. Codebook

The codebook defines each field and its controlled vocabulary, and was revised five times during extraction. The revisions are logged in the repository and are load-bearing:

- **R2** distinguishes a review from an enabling paper, and classifies attack-generating work as *bridge*.
- **R3** requires a named interface or protocol before `attack_surface_named` may be coded *yes*; passing mentions do not qualify.
- **R4** prevents passing mentions from triggering the *multi* lifecycle value.
- **R5** contracts `ml_family` to a controlled vocabulary.
- **R6** fixes evaluation-method precedence (*hardware-in-the-loop* > testbed > simulation) and maps surveys to *concept* maturity.

The `twin_role` field requires positive evidence. *both* needs explicit coverage of the defensive and attack-target roles *and* of the tension between them; *target* requires that the twin be the adversary's objective rather than an asset to be protected. *both* = 0 is therefore definitional: a paper cannot score *both* without stating the tension. Candidates such as [10], "Enhancing the cybersecurity of battery digital twins," were read as *tool* because the twin remains an asset under protection.

4.7. Scoring of AI-threat engagement

Each AI-enabled threat field was coded on a 0–2 ordinal scale: 0 = not addressed; 1 = mentioned, listed, or peripheral (fewer than two sentences or no failure-mode detail); 2 = substantive (a threat model, mechanism, experiment, or mitigation specific to that vector, or a section dedicated to it). The threshold was applied to the full text rather than the abstract. `ai_threat_overall` is 1 if any vector is 1 or 2, and 2 if any vector is 2. Both coders coded the scale independently; per-vector agreement is reported in Section 5.7.

4.8. Reliability

A second coder independently coded a 20% random sample (20 studies) blind to coder 1's values and to the screening label. Agreement is reported field by field in Section 5.7. We use Cohen's κ where it is defined and percentage agreement where a field is near-constant and κ is undefined. With $n = 20$, the large-sample standard error of κ is of the order of 0.2–0.3 even for the better fields; cells in Table 6 are point estimates, and a stable ranking of field difficulty is not achievable from this sample.

4.9. Adjudication

All field-level disagreements were adjudicated against the codebook and the underlying evidence. The rules, stated here rather than buried, are: adopt the value the codebook and evidence dictate; where both coders' values are defensible, coder 1's prevails; where neither fits, the codebook rule decides. Adjudication changed 20 net cells, none of them a headline field (Section 5.7). Because coder-1 precedence is a real limitation of the agreement analysis, we report the raw agreement before adjudication alongside the adjudicated result.

4.10. False-negative audit

Because the headline is an absence—of attack-side engagement—we audited it. We ran a strict AI-threat lexicon over the 48 records coded as *not engaging* any AI threat and a broader eleven-family net. The audit is reported in Section 5.8; it bounds the false-negative channel without eliminating it, and we say so.

4.11. Verification and reproducibility

Every DOI was verified against an authoritative registry: 96 of 97 resolved with a title match (95 exact, one near), one DOI is not registered, and three corpus records carry no DOI. Corpus metadata was cross-checked against Crossref, and 18 of 54 venue labels disagreed, which we resolved in favour of the registry. All extraction data, the codebook, the coding sheets, the reliability script, and the figure script are released as a reproducibility package.

5. Results

5.1. Which denominator supports which claim

Table 2 summarises the denominators behind the headline figures.

Table 2: Denominators used in this review.

Subset	n
Retained corpus	100
Full-text records	83
Abstract-only records	17
ML-used studies	70
Non-review studies	49
Journal articles	62

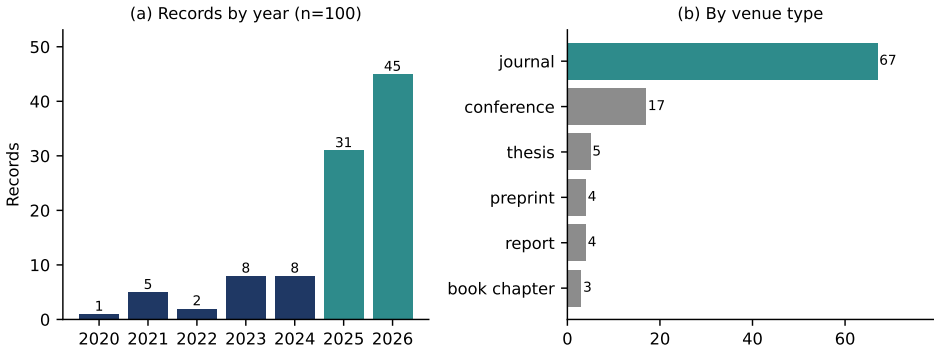


Figure 2: Corpus profile. (a) Records by year. The 2025–2026 concentration reflects selection as much as literature growth, since the top-100 cut favours recency among tied relevance scores. (b) Records by venue type.

5.2. Corpus profile (RQ1, descriptive)

The corpus is 100 studies: 67 journal articles, 17 conference papers, 5 theses, 4 preprints, 4 reports, and 3 book chapters. Publication is recent and accelerating: 1 study from 2020, 5 from 2021, 2 from 2022, 8 from 2023, 8 from 2024, 31 from 2025, and 45 from 2026; 76% of the corpus appeared in 2025 or 2026 (Figure 2a). The leading publishers are MDPI (17), Elsevier (12) and IEEE (8). Table 3 gives the profile.

5.3. How the twin is positioned (RQ1)

Across all 100 records, the adjudicated stance distribution is defensive 42, review 36, bridge 10, enabling 10, adversarial 2; security is the primary subject in 83 of 100 (Figure 3a). The distribution differs from the screening seed—which had adversarial 11, review 23—confirming that keyword-level stance labelling is unreliable and that reading is necessary (Figure 3b).

Table 3: Corpus profile ($n = 100$; full text = 83, abstract-only = 17).

Characteristic	Records
Journal article	67
Conference paper	17
Thesis	5
Preprint	4
Report	4
Book chapter	3
2026	45
2025	31
2024	8
2023	8
2022	2
2021	5
2020	1
Stance: defensive	42
Stance: review	36
Stance: bridge	10
Stance: enabling	10
Stance: adversarial	2

The sharper result is `twin_role`, coded on the 83 full-text records: the twin is treated as a defensive *tool* in 75, as an attack *target* in exactly 1, as *both* in 0, and is *unclear* in 7. No paper in the corpus articulates the dual role that Habeeb et al. name [7], and only one [11] treats the twin as something to be attacked. This is the first half of the asymmetry.

5.4. Threat vocabulary and lifecycle concentration (RQ2)

Table 4 reports the closed threat vocabulary. Denial of service (49), false data injection (46) and malware (34) lead; the AI-relevant terms that the corpus does contain— poisoning (22), evasion (11)—are present but well behind the classical attacks. A mean of 3.77 distinct threats is named per full-text study (median 3, max 11), and 10 of 83 studies name no threat at all, which is itself notable in a security review.

Threat coverage is concentrated almost entirely at the operation phase. Figure 4 crosses the threat vocabulary with `lifecycle_stage`. The design

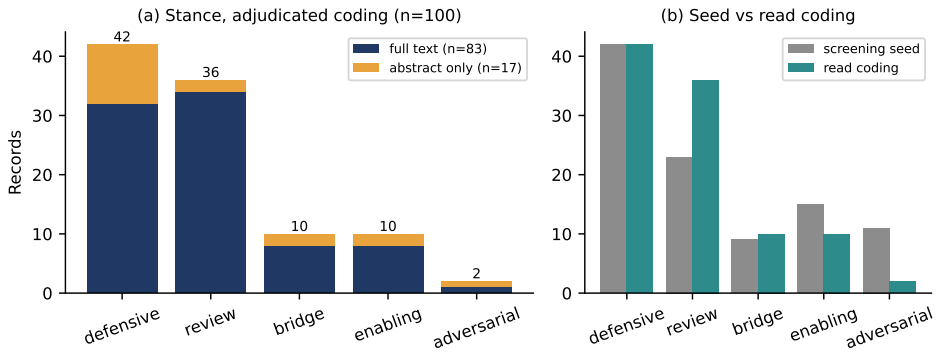


Figure 3: Stance. (a) Adjudicated coding across all 100 records, split by whether full text was coded. (b) Screening seed versus read coding: the shift from *adversarial* to *review*/*defensive* shows the seed over-called an adversarial stance on keyword evidence.

and commissioning columns are effectively empty—one design-phase study [12] and none at commissioning—while the operation column is dense. Across the corpus, lifecycle is operation in 66 of 83 studies (79.5%), *multi* in 12, unclear in 4, design in 1, and commissioning in 0, as coded. Read as single labels, security work concentrates where the twin runs at deployment, not where it is built.

Threat models are also thin. An explicit, stated threat model appears in 19 of 83 studies, and an adaptive adversary is modelled in 10—so the corpus mostly names threats without committing to an attacker model, which is precisely the weakness that makes attack-side claims hard to test.

5.5. Engagement with AI-enabled threats (RQ3)

Note on terminology: the threat-vocabulary count (`poisoning`) and the AI-threat field (`ai_poisoning`) track different constructs; `poisoning` (22 records) records any discussion of poisoning as a grid attack, while `ai_poisoning` (19) records engagement with AI-specific poisoning attacks. This is the headline. Across the 83 full-text records, 35 (42.2%, Wilson interval 32–53%) engage at least one AI-enabled threat at any level, 11 (13.3%, 8–22%) engage one substantively, and 48 (57.8%, 47–68%) do not engage at all (Figure 5). Restricting to the 49 non-review studies changes the counts but not the reading (20 of 49 engage any, 6 of 49 substantively), so the asymmetry is not an artefact of review authorship. Per vector, at any/substantive level: adversarial machine learning 24/8, data or model poisoning 19/7, evasion

Table 4: Threat vocabulary (full text, $n = 83$). Terms are non-exclusive.

Threat	Records
Denial of service (`dos`)	49
False data injection (`fdi`)	46
Malware (`malware`)	34
Man-in-the-middle (`mitm`)	26
Replay (`replay`)	23
False command injection (`false_command`)	22
Data/model poisoning (`poisoning`)	22
Physical attack (`physical`)	21
Insider (`insider`)	20
Supply chain (`supply_chain`)	19
Model evasion (`evasion`)	11
Reverse pathway (`reverse_pathway`)	10
Other	8
Hardware	1
Prompt-level	1
No threat named at all	10
Mean distinct threats per study	3.77

9/1, LLM-specific 4/2, and agentic attack automation 2/1. All 35 engaged studies are among the 70 that use machine learning defensively, so the non-engagement is 35 of 70 (50%) rather than a zero/one split. The asymmetry is real, but it is an asymmetry of emphasis rather than absence.

Two patterns sharpen the result. First, engagement is shallow: mentioned (score 1) is more common than substantive (score 2) for every engaged vector except LLM threats (2 mentioned, 2 substantive, a tie on a base of 4) and agentic threats (1 mentioned, 1 substantive, a tie on a base of 2). Second, the two deepest engagements are reviews [3, 4]; on the review authorship question the substantive set has 5 reviews (45%) against 34 (41%) in the full-text base, i.e. no over- representation; both counts are small and the difference is unsupported by any test.

The abstract-level screening figure is worth stating against this. Over a supplementary 469-record probe pool, an AI-enabled threat appears in the abstract in 24 records (5.1%), while the full-text corpus reads 35 of 83 (42.2%). The two figures are not like-for-like; they are quoted together as

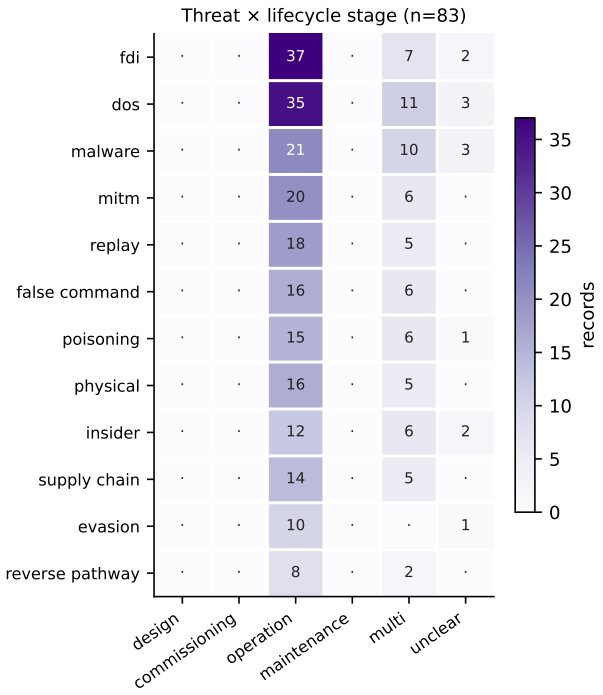


Figure 4: Threat × lifecycle-stage coverage (full text, $n = 83$). Empty cells in the design and commissioning columns are a coding outcome, not a census of practice; the field's agreement is $\kappa = 0.32$.

an illustration of how much screening losses can differ, not as a measured correction factor.

Table 5 lists the eleven substantive studies. Nine are journal articles (all 2025–26), one is a thesis [8] and one a preprint [13] (2022); five carry the *review* stance and three the *bridge* stance, so 8 of 11 are review-or-bridge work. All eleven use ML predominantly for detection. No study's primary contribution is an AI-enabled attack on a grid digital twin; the deepest engagements are themselves reviews [3, 4].

Read qualitatively, the eleven cluster into four patterns. (1) Generative and adversarial modelling as attack generation: the twin is used to synthesise adversarially consistent attacks and GAN-based evasion by the authors themselves [3–5, 9, 13, 14]. (2) Learned intrusion detectors stressed under distribution shift: data poisoning [8, 13, 15], FGSM/JSMA perturbations [4], and GAN-driven evasion [13, 14]. (3) LLM threats treated as architecture-

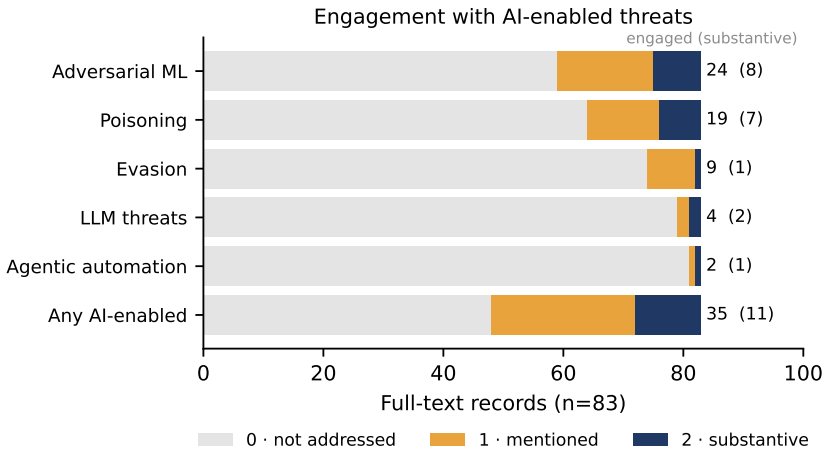


Figure 5: Engagement with AI-enabled threats (full text, $n = 83$). Scale: 0 not addressed, 1 mentioned or peripheral, 2 substantive. The bar labels give engaged (substantive) counts.

level hazards—hallucination, prompt injection, biased recommendations—in the review literature [3–5]. (4) Agentic automation represented only once, in [4]; no study constructs an adversarial RL or agentic pipeline. "Substantive engage- ment" therefore currently means backgrounding a threat while retaining a defensive detector.

5.6. Evidence quality (RQ4)

Machine learning is used in 70 of 83 full-text studies; its dominant role is detection (55), followed by forecasting (12), none (13), and control (3). Evaluation method splits as simulation 33, qualitative 28, testbed 10, conceptual 5, mixed 5, and hardware-in-the-loop 2 ([16, 17] after adjudication). Maturity is prototype in 46, concept in 34, and validated in only 3. Baselines are reported in 41 of 83 studies and quantitative metrics in 43, while limitations are stated in 67. Code is released in 3 of 83 studies and data in 16. The evidence base is, in short, prototyping-heavy and release-light. A venue-quality sensitivity (restricting to the 62 journal articles) leaves the asymmetry intact (26/62 any; 9/62 substantive; 36/62 none; 58/62 tool; twin targeted by one), though it discards roughly a quarter of the corpus. Restricting the denominators to the 49 non-review studies leaves the engagement result similar (20/49 any versus 35/83) but not the baseline result: baselines rise from 41/83 (49%) to 37/49 (76%), and metrics from 43/83 (52%) to 40/49 (82%).

Table 5: The eleven studies engaging an AI-enabled threat substantively (score 2). IDs are corpus record numbers, not references. Vectors are non-exclusive. Qualitative pattern markers are in the prose above.

ID	Venue (type)	Year	Stance	AI-threat vectors
16	Energies (journal)	2026	review	adv. ML, poisoning, LLM
17	Sustain. Energy Technol. Assess. (journal)	2026	defensive	poisoning
22	Informatics and Automation (journal)	2026	bridge	adv. ML
24	Electronics (journal)	2026	review	adv. ML, poisoning
49	Sensors (journal)	2025	bridge	adv. ML
50	Energy Convers. Manag. X (journal)	2025	review	all five vectors
54	Energies (journal)	2025	defensive	adv. ML
57	Energies (journal)	2025	review	poisoning
69	Int. Res. J. Modernization (journal)	2025	bridge	adv. ML
74	Deep Blue, Univ. Michigan (thesis)	2025	defensive	poisoning
94	arXiv (preprint)	2022	review	adv. ML, poisoning

Code and data remain low in both cases.

Grid domains are dominated by studies that span multiple domains (*multi*, 47). Where a single domain is named: distribution 11, unclear 7, EV charging 6, microgrid 6, transmission 5, and substation 1. Transmission—often assumed to be the security-critical domain—is under-represented relative to distribution and EV charging.

5.7. Reliability results

Table 6 reports agreement on the 20% sample. On `ai_threat_overall`, the 3x3 counts are (rows coder 1, columns coder 2): 0: 9, 1, 1; 1: 1, 3, 1; 2: 1, 2, 1. Collapsing to "any engagement vs. none": coder 1 has 9 of 20, coder 2 has 9 of 20, agreement 16/20 ($\kappa = 0.60$). Collapsing to "substantive vs. not": agreement 15/20, but $\kappa = 0.14$, because only 1 of the 6 studies is labelled substantively by both coders (the single (2,2) cell). Under an asymptotic interval on this n = 20 sample, the 95% interval is [-0.52, 0.79] for the substantive collapse and [0.24, 0.95] for the any-engagement collapse; the two are not comparably reliable. This paragraph is a post-hoc reading of the 3x3 once we saw it, and is flagged as such. The disagreement on "any engagement" is symmetric (2 each way), so there is no evidence that 35/83 is an undercount; the headline supported by the data is the 35/83 any, not the 11/83 substantive, which is fragile.

Fields with substantial agreement include `baseline_compared` ($\kappa = 0.90$), `twin_role` (0.77), `ai_poisoning` (0.73), `threat_model_explicit` (0.69) and `metrics_reported` (0.68). Fields at moderate or fair agreement include the ones the headline most needs: `ai_threat_overall` (0.41), `ai_evasion` (0.39), and the threat-vocabulary fields. `attack_surface_named` is the weakest (0.14). Where a field is near-constant, κ is undefined and we report percentage agreement instead: `stance_confidence`, `limitations_stated`, `data_available`, `ai_llm`, `ai_agentic` and `code_available`.

We report these numbers because they bound the result. The headline's core count—whether a study engages any AI-enabled threat—is coded at only moderate agreement (0.41), so the 35/83 figure carries real uncertainty. Two things mitigate it: adjudication moved no headline field, and the false-negative audit (Section 5.8) finds the absence robust under two independent lexicons. But the honest reading is that the asymmetry's *direction* is well supported while its *exact magnitude* should be quoted with care, and we do.

A systematic style difference between coders emerged and is worth recording as a codebook problem rather than a coder problem: coder 1 tends to assign the default category (*high*, *yes*, *operation*, *single*) whereas coder 2 differentiates (*medium*, *no*, *multi*, *unclear*) and names more attack surfaces. Where this is a default-category artefact, it depresses κ without indicating a substantive disagreement.

5.8. False-negative audit

The strict AI-threat lexicon flagged none of the 48 records coded as not engaging, which is consistent with the coding. But the audit is not, on its own, confirmatory: as a positive control the same lexicon recovers only 7 of 11 known-positive studies, because some positives use generic vocabulary the lexicon lacks (e.g. [9, 15, 18, 19]). A pattern set cannot recognise wording it was not written for.

We therefore also ran a broader eleven-family net. Of the 48 negatives, 38 fired on some family and 23 fired on a strong family. Inspection shows all 23 to be either conventional cyberattacks discussed without any AI component or known false positives. The instructive false positives justify human coding over keyword coding: `GaN` (the semiconductor) matching "GAN" in two records [10, 20], the author surname "Gan" in another [21], Xenon-135 reactor physics matching "poisoning" [22], ARP poisoning [23], and the SolarWinds incident discussed as a data-poisoning example [24]. Two close calls were examined and cleared: [25] bypasses *classical* state-estimation bad-data

detection, not an ML detector, and [26] concerns a SCADA supervisor with AI detection stated only as future work. No codes were changed. The audit bounds but does not eliminate the false-negative channel, because no lexicon matches wording it does not contain.

5.9. Adjudication outcome

Adjudication resolved 149 field-level disagreements and changed 20 net cells from coder 1's values: `ml_family` (12), `threat_models_named` (2), `stance` (2), `eval_method` (2), `dt_type` (1) and `grid_domain` (1). Three substantive corrections came out of it: two studies move to *bridge* stance [14, 18], and one study is a substation-domain study [17], contrary to its initial coding. Two third-value resolutions occurred, both `eval_method` = hardware-in-the-loop [16, 17]. No headline field moved, which is the point: the central result is not an artefact of the tie-break rule.

6. Discussion

6.1. The asymmetry, quantified

The corpus adopts AI defensively in 70 of 83 full-text studies and engages AI-enabled threats in 35, substantively in 11. That gap—between near-universal defensive use and minority attack-side engagement—is the paper's central finding, and it is robust to adjudication and to the false-negative audit in direction if not in exact size. The practical reading is that the field has built a large defensive apparatus on models whose own failure modes it rarely studies.

6.2. The twin as a target is thinly evidenced

As it stands, only one of our 83 full-text studies treats the twin as the adversary's objective [11], and it concerns vehicle twins rather than grids. Habeeb et al. name the twin as an expanded attack surface [7]; we cannot speak to that broader version with a coding rule defined as "the adversary's objective," so the grid-side evidence for attack-surface engagement is, on these definitions, all but absent.

6.3. The lifecycle blind spot is the deeper problem

The result with the longest reach in the data, if it holds, is the lifecycle concentration: 66 of 83 studies are coded at the operation phase, one at design, none at commissioning. Two caveats apply. First, the field carries our lowest inter-coder agreement ($\kappa = 0.32$), and coder 1, who coded 80% of the corpus, defaults to 'operation'; the 12 studies coded 'multi' may contain design or commissioning activity that the single label suppresses. Second, the score records the phase a study *targets*, not the phases it necessarily *touches*. Read conservatively, the finding is therefore that the literature is dominated by studies of running twins, with almost no study that sets out to secure a twin before it is built, not that no design-phase security work exists.

6.4. Why reviews are where the threats are named

The two deepest engagements are reviews [3, 4]; 5 of the 11 substantive studies (45%) are reviews against 34 of 83 (41%) in the full-text corpus, so review authorship is not over-represented, but bridge studies are (3 of 11 against 8 of 83 full-text records, 2 more in the abstract-only set). The honest reading is that reviews have the length and the licence to survey AI threats, while primary papers have neither. Document length is a clear confound and these numbers should not be read as evidence that 'only reviewers care.' Also, the abstract-versus-full-text comparison in Section 5.8 is not like-for-like (24/469 comes from a wider query set and a different threshold than 35/83) and should be read as an illustration of screening loss rather than an 8x correction factor.

6.5. Positioning against the prior review

Relative to Habeeb et al. [7], our contribution is a grid-side probe of a narrower instance of one of their axes, with a published corpus, a closed vocabulary, and reported reliability; full scope alignment is in Section 6.2. Table 1 shows that our independent vocabulary covers all five of their vectors, so the comparison is not competing but additive. Two caveats are structural, not rhetorical. First, their search string has no specific attack-side operator, so records enter only through security terms; ours adds four adversarial-AI probes and still finds shallow engagement, so the gap is not merely their query's doing. Second, their corpus spans manufacturing, healthcare, water and smart city, where the digital twin's role differs from the grid's bidirectional, safety-critical loop; grid conclusions should not be generalised from it, nor it from us.

6.6. Operational reading

For operators, three interpretive notes follow (these are author readings, not coded evidence). First, a twin whose learner is trained on the same telemetry it is meant to defend has no independent ground truth and is a single point of failure that the corpus rarely names. Second, runtime detection at the operation phase is necessary, but cannot cover design-stage trust failures. Third, the near-total absence of released code and data means a twin procurement decision cannot currently be informed by reproducible attack evidence.

7. Research agenda

We state five items. Each is tied to the gap that justifies it (from Section 5) and to the validation that would confirm it, so the agenda is testable rather than aspirational. Table 7 summarises.

P1 — Measure the twin as an attack target, not only as a defence. Justified by `twin_role` target = 1; validated by a reproducible attack against a grid twin where the twin is the objective, with released code and a stated adaptive adversary.

P2 — Move security to design and commissioning. Justified by operation = 66, design = 1, commissioning = 0; validated by twins whose trust boundaries, interface contracts and fail-safe behaviour are specified and tested before deployment.

P3 — Adopt explicit adaptive threat models. Justified by an explicit threat model in 19/83 and an adaptive adversary in 10/83; validated by studies that state the attacker's knowledge, capability and objective and test a reactive attacker.

P4 — Defend the learned components, not just the plant. Justified by detection-dominant ML use (55/83) against shallow poisoning and evasion engagement (22 and 11); validated by poisoning- and evasion-robust twins under distribution shift.

P5 — Release code and data. Justified by code in 3/83 (3/49 non-review) and data in 16/83; baseline disclosure alone is high among primary studies, but reproducibility remains impossible without code or data.

P5 must be stated carefully. We measure *research attention*, not *deployment security*. A low attack-side count does not mean grids are undefended; it means the published research has not modelled these attacks. We cannot estimate either the probability of an AI-enabled attack or its expected impact from this corpus, and we do not.

8. Threats to validity

8.1. Construct validity

"AI-enabled threat" is a construct, and the false-negative audit shows a strict lexicon misses some positives (7/11 recovery). We mitigated by human coding with context, by the broad eleven-family net, and by the conservative `twin_role` rule, but the construct remains softer than the threat-vocabulary fields.

8.2. Internal validity and screening

Screening on title and abstract is imperfect: in the probes built into this study, an AI-threat term appears in the abstract in 24 of 469 (5.1%) records, whereas the retained full-text corpus reads 35 of 83 (42.2%). The two figures use different query sets and different screening thresholds, so they are an illustration of screening loss, not a like-for-like comparison, and the 'order of magnitude' distance between them should not be quoted.

8.3. Reliability

Agreement on headline fields is moderate, not strong (`ai_threat_overall` $\kappa = 0.41$; `attack_surface_named` 0.14). Coder-1 precedence in adjudication is a real limitation and is declared. The exact magnitude of the asymmetry should therefore be read with the reliability table, not in isolation.

8.4. External validity and coverage

PRISMA compliance is partial: one database, no stated search date in the text, no registered protocol and five codebook revisions after the protocol was 'fixed'—all of which limit how strongly the review answers the protocol hypothesis.

Seventeen of the 100 records were available only as abstracts, hence the corpus has a 17% non-full-text rate; those records are listed with DOIs so readers can check what is missing. One corpus DOI does not resolve (record

18, which lies within the reliability sample) and three records have no DOI; these are disclosed rather than dropped. Venue labels disagreed with Crossref in 18 of 54 cases and were corrected in favour of the registry.

8.5. Search bias

The core grid-security vocabulary does not retrieve attack-side work; the four adversarial-AI probes were added to widen the candidate pool (approximately 2,000 records), and the released files do not yet separate probe-only membership in the final 100; this is a gap in the record. Even with them, engagement is shallow, so the asymmetry is not merely a retrieval artefact—but the corpus should still be read as a bounded sample rather than a census. The recency component of the ranking also means foundational attack-side work from 2020–2023 may be under-weighted.

8.6. Researcher degrees of freedom

Five codebook revisions during extraction are a real degree of freedom. They are logged, dated, and applied consistently to all 100 records, and adjudication used the same rules; but a reader who distrusts post-hoc refinement should weight the raw pre-adjudication codes, which are released frozen alongside the adjudicated matrix.

8.7. Quality appraisal, venue quality and scope

At least one retained record [27] carries anonymous authorship in a low-tier journal, which is exactly why the venue sensitivity and DOI verification passes above were necessary. The corpus contains 67 journal articles, 17 conference papers, 5 theses, 4 preprints, 4 reports and 3 book chapters, drawn from venues of varying strictness (including IRJMETS and GSC Online Press). No formal risk-of-bias or venue-quality appraisal was performed; all prevalence figures are therefore reported over the full corpus, and Section 5 reports sensitivity subsets for non-review studies. Readers should weight venue quality into any claim reproduced from this review. The claim in Section 1 is restricted to the intersection of 'digital twin' and 'cybersecurity' in power grids; substantial adversarial-ML work on grid state estimation lies outside that intersection and is not covered by the headline numbers.

9. Conclusion

Grid digital-twin security has adopted AI as a defensive tool far faster than it has modelled AI—or the twin itself—as an attack target. Across a screened corpus of 100 studies, 76% published in 2025–26, AI is defensive in 70 of 83 full-text studies while 35 engage any AI-enabled threat and 11 engage one substantively; the twin is an explicit attack target in exactly one study. Coverage is dominated by the operation phase (66 of 83 coded as such, on a field with $\kappa = 0.32$); at most, one study is coded at design and none at commissioning, and threat models are thin (explicit in 19/83). These directions are robust to adjudication and to a two-lexicon false-negative audit, though the exact magnitudes are bounded by moderate coder agreement on the headline fields. The closest prior review names the dual role; this review tests one of its axes at grid scale and finds it largely empty. The five-item agenda that follows is deliberately testable: each item names the gap that justifies it and the evidence that would close it.

Highlights

- 100-study PRISMA-2020-style corpus on digital twins for power-grid cybersecurity (2020–2026).
- 70/83 defensive, 35/83 engage an AI-enabled threat, 11/83 substantively (fragile, post-hoc); the twin is an explicit target in at most one study (a vehicle-twin review).
- Coverage, as coded: 66/83 at operation, 1 at design, 0 at commissioning.
- Five-item agenda, each tied to its gap and its validating evidence.

Declaration of competing interest

The authors declare no competing interests.

Data availability

All extraction data, the codebook, the coder sheets, the reliability script and the figure script are available in the reproducibility package accompanying this review (repository link supplied at submission).

CRediT authorship contribution statement

Author One: conceptualisation, methodology, formal analysis, writing—original draft. **Author Two:** validation, reliability coding, writing—review and editing.

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Table 6: Inter-coder agreement on the 20% blind sample ($n = 20$). κ is reported where interpretable; † marks near-constant fields (modal category $\geq 90\%$ in coder 1) where κ is not interpretable and percentage agreement is reported instead.

Field	% agree	κ / set metric
Cohen's κ interpretable		
baseline_compared	95	0.90
twin_role	95	0.77
ai_poisoning	90	0.73
threat_model_explicit	90	0.69
metrics_reported	85	0.68
security_primary	85	0.58
ml_used	85	0.57
ml_role	70	0.55
stance	60	0.46
dt_type	75	0.46
grid_domain	65	0.44
ai_adv_ml	70	0.44
maturity	60	0.42
ai_threat_overall	**65**	**0.41**
ai_evasion	75	0.39
eval_method	50	0.37
lifecycle_stage	65	0.32
attack_surface_named	60	0.14
Percentage agreement only (†)		
code_available	100	†
ai_agentic	95	†
ai_llm	90	†
data_available	85	†
stance_confidence	70	†
limitations_stated	70	†
Multi-label fields (set metrics)		
threat_models_named	5	Jaccard 0.43
ml_family	20	Jaccard 0.42
threat_count	10	mean $\lvert\Delta\rvert$ 1.80

Table 7: Research agenda: gap, direction, and the validation each item demands.

#	Gap (evidence)	Direction	Validation
P1	Twin as target: 1/83	Treat the twin as the adversary's objective	Reproducible twin-directed attack with released code, adaptive adversary
P2	Design 1/83, commissioning 0/83	Secure design and commissioning phases	Trust boundaries and failsafes specified and tested predeployment
P3	Explicit threat model 19/83; adaptive 10/83	State and test adaptive threat models	Attacker knowledge, capability, objective stated; reactive attacker tested
P4	Poisoning 19, evasion 9 vs detection 55	Defend learned components	Poisoning/evasion-robust twin under distribution shift
P5	Code 3/83 (3/49 non-review), data 16/83	Release artifacts	Third party reruns unaided